



Development of Universal Hydraulic System

Project Details:
Project Value:
EDC:

DPP2204001-2
₹1,20,00,000/- + GST 18%
31st Dec 2024

Over view:

This facility creation is aimed at creating a common hydraulic fluid power source for all the test stations housed at CMTI, and also enhance the current limitations of the test systems in terms of pressure, flow and temperature control.

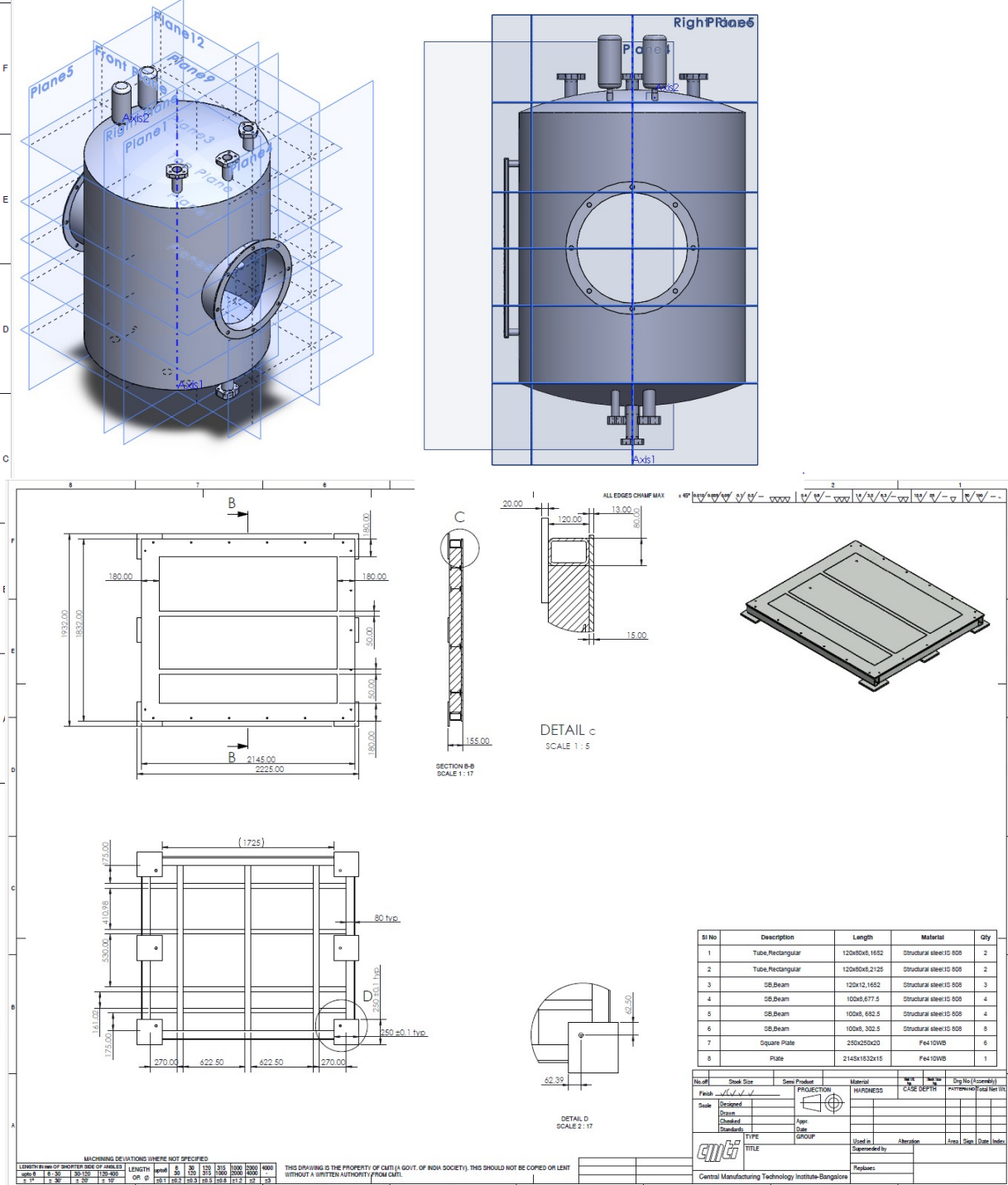
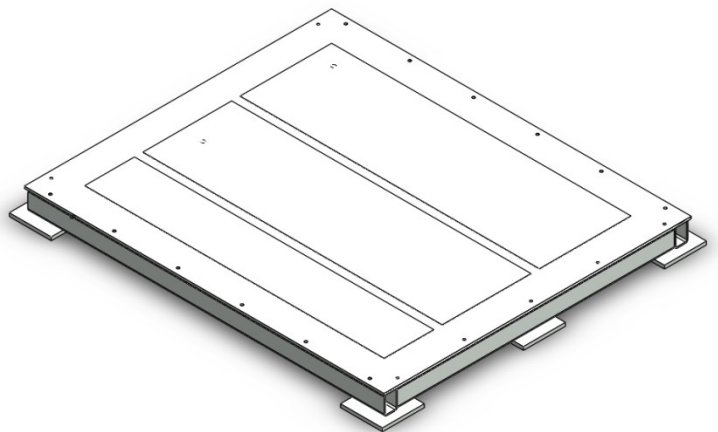
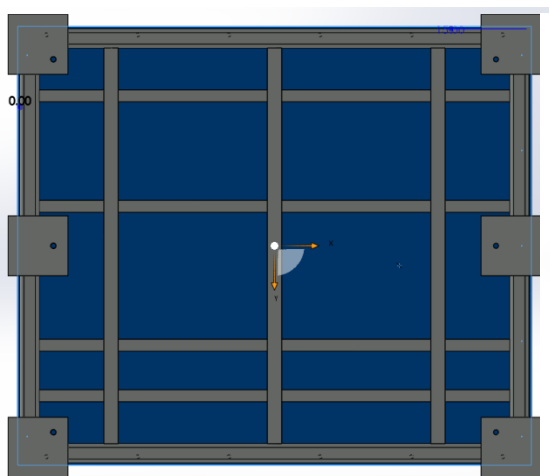
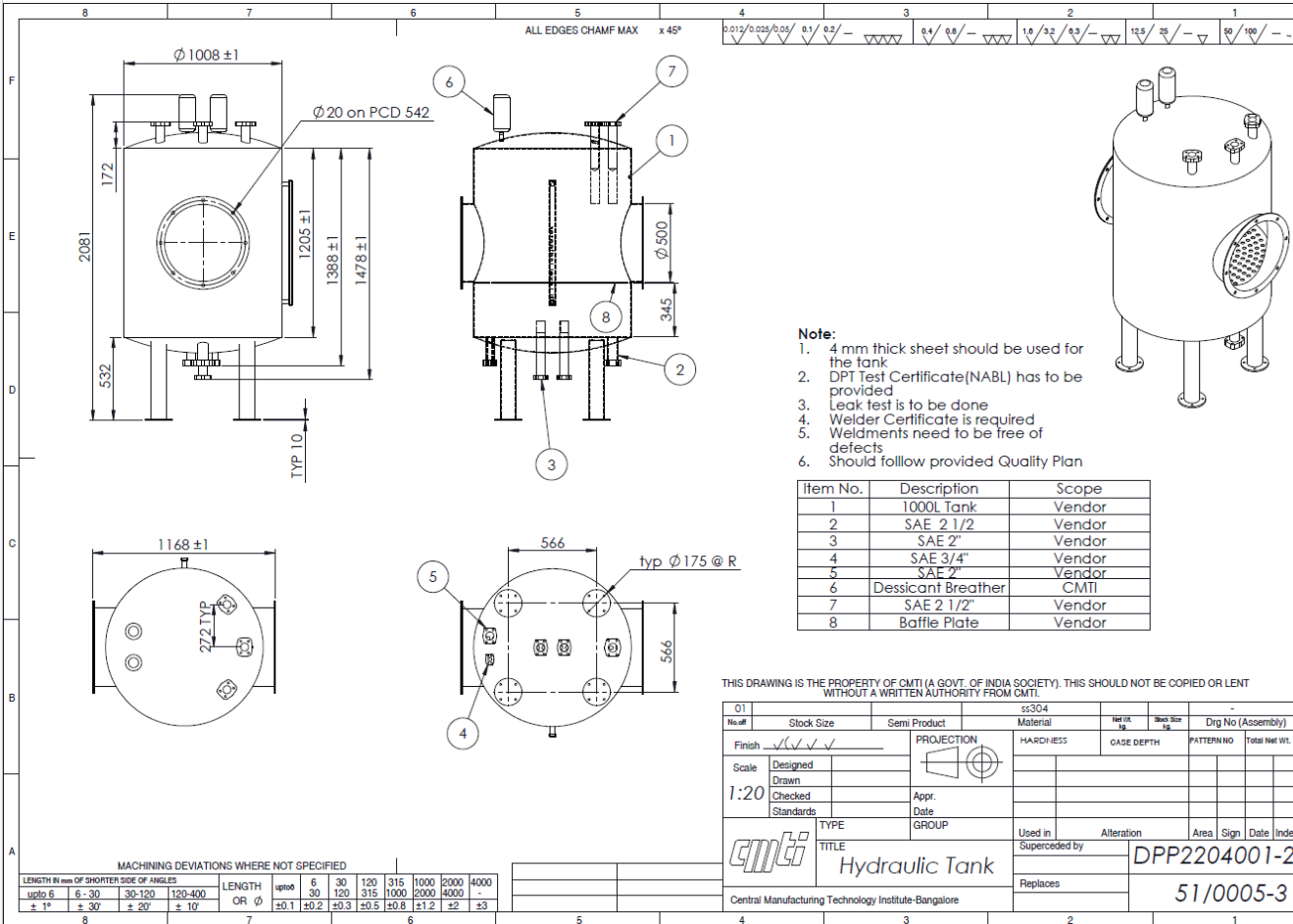
The system consists of two stations, each consisting of a 132kW AC synchronous motor with VFD for shaft speed control, coupled to a bent axis axial piston pump which can provide 150lpm at 350bar. These identical systems will have provision for remote pressure and flow control to suit the test requirements.

The hydraulic oil for these systems are sourced from a 1000L SS304 hydraulic reservoir with suction strainers and baffle plates. Filtration system consists of filters ranging from $3\mu\text{m}$ to $10\mu\text{m}$. An oil chiller of 30TR is provided for effective temperature control of the hydraulic oil as the heat load from the pumps are up to 110kW.



Bill of materials with status (major)

SI no	Description	Specification	QTY	Make	Status	EDC
1	Oil Chiller	20TR	01 no	Hi-tech cooling solutions	Received	NA
2	Variable frequency drives	160kW	02 nos	Schneider	Received	NA
3	Pilot pump - motor set	---	01 no	Rexroth	Received	NA
3	Oil filters (suction line, return line and pressure line filters)	--	02 nos each	MP filtri	PO placed	31/7/24
4	Flowmeter	Screw type	03 nos	VSE	PO placed	1/9/24
5	Hydraulic oil	MIL-PRF-5606H	1240 litre	Generic	PO placed	1/9/24 (1 month from date of PO)
6	Pump- Motor assembly	150lpm, 350bar, 132kW	02 nos	TBD	Tendering activity is in process	2months from date of PO (15/10/2024)
7	Base frame and skid	1.8mx3m	01 no	Custom built	Tendering activity is in process	2-3 weeks from date of PO (10/8/2024)
				Custom built	Tendering activity is in process	2-3 weeks from date of PO



Booster pump-
motor set
20 TR (60,000 kCal/hr) oil chiller





Variable frequency
drives
(160kW)



Brazed type - plate heat exchangers
(30kW) each.



CNP make- Centrifugal pump: 4m³/hr
(67lpm); 31m; 2900rpm.



CNP make- Centrifugal pump: 10m³/hr
(167lpm); 31m; 2900rpm.

CNP make- Centrifugal pump: 2m³/hr
(30lpm); 110m; 2900rpm.

